

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electronics and Electrical Communication Engineering

2. Subject Name: Digital Signal Processing- I L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: Signals and Systems(EC21208)

4. Syllabus and reference books:

Syllabus:

Review of discrete time signals and systems - causality, stability, discrete time Fourier transform, sampling and z-transform. Introduction to signal space, orthogonal basis and signal representation. Properties of Discrete Fourier Transform (DFT), convolution, Fast Fourier Transform (FFT). Structures for Discrete-time systems, Digital filter design techniques, transformation of digital filters. Finite-word length effects, Quantization. Random Signals and its processing, Spectral analysis, Optimum Linear filters, Prediction, Autoregressive and parametric signal modelling.

Reference Books:

1. Discrete-time Signal Processing: Alan V. Oppenheim and Ronald W. Schaffer, Prentice Hall, 1999.
2. Digital Signal Processing: Principles, Algorithms, and Applications: John G. Proakis and Dimitris G. Manolakis, Cambridge University Press, 2011.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Review of discrete time signals and systems - causality, stability, discrete time Fourier transform, sampling and z-transform.	6
2.	Introduction to signal space, orthogonal basis and signal representation using unitary transforms like DFT, DCT, Haar and Walsh Hadamard transform.	8
3.	Properties of Discrete Fourier Transform (DFT), circular convolution, linear convolution using DFT, overlap add and save methods, Fast Fourier Transform (FFT).	6
4.	Structures for Discrete-time systems, signal-flow graphs,	4

	Filter structures for IIR and FIR filters, lattice structures.	
5.	Digital filter design techniques, FIR filter design, IIR filter design, transformation of digital filters	6
6.	Finite-word length effects: Quantization, Statistical analysis of quantization error, Oversampling A/D and D/A conversion, Quantization of filter coefficients, Effects of finite word-length on digital filters.	4
7.	Random Signals and its processing: Discrete random signals, random process, covariance, correlation, linear estimation, Spectral analysis	6
8.	Optimum Linear filters, Prediction, Autoregressive and parametric signal modelling, Least squares, Prony's method, Spectrum Estimation	8
Total number of hours		48